Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
Student Name	N. Tejasri
Roll Number	2520030132
Branch	CSE
Course Outcome	5

Case Study Topic: Digital Voting & Election Analytics System using Heap Sort

Work Done Summary:

Implemented **Heap Sort** in Java for election analytics and result reporting. Modeled candidates and their vote counts as records stored in an array. Heap Sort was used to organize candidates based on total votes received and generate ranked election results efficiently. Implemented voter registration, authentication verification, ballot casting, vote counting, election analytics, and fraud detection using sorting techniques. Demonstrated sorting of candidate vote counts to identify winners and generate election rankings. Compared Heap Sort with Bubble Sort – Heap Sort is more efficient for large election datasets due to its O(n log n) performance and in-place sorting capability

Implementation Details :

1. Created Candidate class with fields: candidateId, candidateName, and voteCount.
2. Stored candidate records in an array.
3. Implemented heapify() function to maintain max-heap property.
4. Built a max heap using candidate vote counts as keys.
5. Performed Heap Sort to arrange candidates according to vote counts.
6. Generated election rankings after sorting.
7. Fraud detection performed by checking abnormal vote count entries before sorting.
8. Main method inserted candidate data and demonstrated Heap Sort based election result analysis.

Result : Screenshots/Output

```
PS D:\JavaPrograms> javac SecureVoteHeapSort.java
PS D:\JavaPrograms> java SecureVoteHeapSort
SecureVote - Election Analytics System
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Before Heap Sort:
101  Candidate A  450 votes
102  Candidate B  620 votes
103  Candidate C  300 votes
104  Candidate D  780 votes
105  Candidate E  510 votes

Fraud Detection Check:
No Fraudulent Activity Detected.

Election Results After Heap Sort:
103  Candidate C  300 votes
101  Candidate A  450 votes
105  Candidate E  510 votes
102  Candidate B  620 votes
104  Candidate D  780 votes

Winner:
Candidate D with 780 votes
```

Time Complexity:

- Heap Construction: **O(n)**
- Heap Sort: **O(n log n)**
- Heap Sort efficiently ranks candidates and generates election reports for large-scale voting systems.

GitHub Link : https://github.com/tejasri3123/Dsa-2

Student Signature	Faculty Signature